

Set-1

Create following tables with primary and foreign keys and solve the queries given below. Primary keys are underlined

Person (driver_id, name, address)

Car (license, model, year)

Accident (report_no, date_acc, location)

Owns (driver_id, license)

Participated (driver_id, model, report_no, damage_amount)

- 1) Insert 5 records into each table
- 2) Display the Schema of each table.
- 3) Display all records from the person table
- 4) Set a **Foreign key** constraint from Participated.report_no to Accident.report_no.
- 5) Add check constraints on Damage_amount that amt should not be negative.
- 6) Write a query to make the name column in the Person table **NOT NULL**.
- 7) Rename the Car table to Vehicle.
- 8) Create a new table named new_person from existing person table and drop the table new_person.

Set 2

Create Customer and Account table and add rows shown below

Customer Table

C_Id	Cname	City
1	John	Nashik
2	Seema	Aurangabad
3	Amita	Nagar
4	Rakesh	Pune
5	Samata	Nashik
6	Ankita	Chandwad
7	Bhavika	Pune
8	Deepa	Mumbai
9	Nitin	Nagpur
10	Pooja	Pune

Account Table

C_Id	Acc_Type	Amount
1	Current	5000
2	Saving	20000
3	Saving	70000
4	Saving	50000
6	Current	35000
7	Loan	30000
8	Saving	50000
9	Saving	90000
10	Loan	8000
11	Current	45000

- 1) Show the cname, Acc_Type, amount information of customer who is having an saving account.
2. Display the data using Natural, left and right join.
3. Display the information of customers living in the same city as of 'pooja'.
4. Display the information of account, having less amount than average amount throughout the bank.
5. Display the C_id having maximum amount in account.
6. Display the amount and acc_type of those customers whose amount is the minimum amount of that Acc_type.
7. Display the amount of those accounts whose amount is higher than amount of any saving account amount.

Set 3

Create Employee and Department table by identifying Primary key and foreign keys and solve the following queries

Employee (emp_id, emp_name, city, salary, department)

Department (dept_id, dept_name, location)

1. Display schema of each table in each table.
2. Insert 10 records in each table
3. Insert a new department into the Department table.
4. Insert an employee record without specifying the salary column.
5. Update the salary of all employees by adding 20% increment.
6. Display employee name and annual salary of employees whose names have the second letter e.
7. Update the city to 'Pune' for employees working in the 'IT' department.

Set 4

Create Employee and Department table by identifying Primary key and foreign keys

Employee (emp_id, emp_name, city, salary, department)

Department (dept_id, dept_name, location)

1. Display schema of each table.
2. Insert 10 records in each table
3. Update the city to 'Pune' for employees working in the 'IT' department.
4. Delete employee records whose salary is greater than 100000.
5. Display employee name and salary after deducting 5% tax.
6. Display employees who do not live in 'Mumbai' and whose names start with a.
7. Display employees who belong to 'IT' department and have salary greater than 50000.

Set 5

1. Create a new user named with Class_div_rollno.
2. Grant SELECT permission on the Employee table to a user named Class_div_rollno
3. Grant INSERT and UPDATE permissions on the Employee table to Class_div_rollno
4. Revoke UPDATE permission from user1.
5. Create a role named manager_role.
6. Grant SELECT and DELETE privileges on Employee table to manager_role.
7. Revoke the role manager_role from user2

Set 6

Create the tables below and solve the following queries.

Employees

Name	Null?	Type

EMPLOYEE_ID	NOT NULL	NUMBER(6)
FIRST_NAME		VARCHAR2(20)
LAST_NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(25)
PHONE_NUMBER		VARCHAR2(20)
HIRE_DATE	NOT NULL	DATE
JOB_ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION_PCT		NUMBER(2,2)
MANAGER_ID		NUMBER(6)
DEPARTMENT_ID		NUMBER(4)

Departments

Name	Null?	Type

DEPARTMENT_ID	NOT NULL	NUMBER(4)
DEPARTMENT_NAME	NOT NULL	VARCHAR2(30)
MANAGER_ID		NUMBER(6)
LOCATION_ID		NUMBER(4)

1. Create EmpTest from Emp table by copying structure and data.
2. Add a new row into the EmpTest table for Empno,Ename,Sal columns. In Ename should have the Current User.
3. Update EmpTest by increasing the Salary of Vaishali by 15%. Confirm your changes.
4. Increase all the employees salary by 10% in EmpTest table who are working in PUNE
5. Delete all the Comm data from EmpTest Table.
6. Delete all the employees in deptno 10 and do not issue a Commit.
7. Issue a Rollback and reflect changes.

Set 7

Create the table below and solve the following queries.

Employees

Name	Null?	Type

EMPLOYEE_ID	NOT NULL	NUMBER(6)
FIRST_NAME		VARCHAR2(20)
LAST_NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(25)
PHONE_NUMBER		VARCHAR2(20)
HIRE_DATE	NOT NULL	DATE
JOB_ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION_PCT		NUMBER(2,2)
MANAGER_ID		NUMBER(6)
DEPARTMENT_ID		NUMBER(4)

1. Start a transaction and insert a new employee record. Commit the transaction.
2. Start a transaction, update employee salary, and rollback the transaction.
3. Start a transaction, create a savepoint named SP1, update salary, and rollback to SP1.
4. Demonstrate the use of multiple savepoints within a single transaction.
5. Start a transaction, delete an employee record, and commit the changes.

Set 8

Create the employee table with all constraints.

Name	Null?	Type

EMPLOYEE_ID	NOT NULL	NUMBER(6)
FIRST_NAME		VARCHAR2(20)
LAST_NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(25)
PHONE_NUMBER		VARCHAR2(20)
HIRE_DATE	NOT NULL	DATE
JOB_ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION_PCT		NUMBER(2,2)
MANAGER_ID		NUMBER(6)
DEPARTMENT_ID		NUMBER(4)

1. Find the highest, lowest, sum and average salary of all employees. Label the columns.
2. write a query to display the number of people with the same job.
3. Determine the number of managers without listing them. Label the column number of Managers
4. Find the difference between the highest and lowest salaries. Label the column DIFFERENCE.
5. Create a report to display the manager number and the salary of the lowest-paid employee for that manager.

Set 9

Create the employee table with all constraints and solve the following queries. -

Employee (emp_id, emp_name, department, city, salary)

1. Display department-wise total salary of employees.
2. Display department-wise average salary.
3. City-wise employee count greater than 3
4. Department-wise maximum salary above 60,000
5. Display department and city-wise total salary.
6. Display departments having more than one employee.
7. Display city-wise employee count in descending order.

Set 10

Create the employee table with all constraints and solve the following queries. -

Employee (emp_id, emp_name, department, city, salary)

1. Department-wise total salary greater than 2,00,000
2. Department-wise average salary more than 40,000
3. Display city-wise number of employees.
4. Display department-wise maximum salary.
5. Display city-wise average salary.
6. Display department-wise employee count where the number of employees is more than 3.
7. Display city-wise employee count in ascending order.

Set 11

1. Create a table dept1 as below with following columns:

DEPTNO integer pk, dname varchar2(30) not null, loc varchar2(30) not null

2. Add column mgr as number in the dept table.

3. Create table empl with following columns

empno integer, ename varchar2(20), sal number(10,2), deptno number

4. Add constraints on empl table

Empno-primary key

Ename-not null

Check sal>10000

Deptno as foreign key

5. Create table dept11, emp11 from dept1, empl

6. Add a new column address varchar2(30) in the empl table

7. Rename sal column in empl table to salary

8. Display constraints names and verify the names for the empl

Set 12

Create the tables below and solve the following queries.

Employees

Name	Null?	Type

EMPLOYEE_ID	NOT NULL	NUMBER(6)
FIRST_NAME		VARCHAR2(20)
LAST_NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(25)
PHONE_NUMBER		VARCHAR2(20)
HIRE_DATE	NOT NULL	DATE
JOB_ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION_PCT		NUMBER(2,2)
MANAGER_ID		NUMBER(6)
DEPARTMENT_ID		NUMBER(4)

Departments

Name	Null?	Type

DEPARTMENT_ID	NOT NULL	NUMBER(4)
DEPARTMENT_NAME	NOT NULL	VARCHAR2(30)
MANAGER_ID		NUMBER(6)
LOCATION_ID		NUMBER(4)

1. Write a query for the HR department to produce the addresses of all the departments. Use the EMP and DEPT tables. Show the EMPNO, ENAME, SAL, DNAME and LOC in the output. Use a NATURAL JOIN to produce the results.
2. The HR department needs a report of all employees. Write a query to display the JOB, MGR, SAL, COMM, DNAME of employees whose JOB is SALESMAN.
3. The HR department needs a report of employees in LOC DALLAS. Display the ENAME, job, DEPTNO and DNAME for all employees who work in DALLAS.
4. Modify the previous Query to display all employees including King, who has no manager. Order the results by the employee number.

Set 13

Employees

Name	Null?	Type

EMPLOYEE_ID	NOT NULL	NUMBER(6)
FIRST_NAME		VARCHAR2(20)
LAST_NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(25)
PHONE_NUMBER		VARCHAR2(20)
HIRE_DATE	NOT NULL	DATE
JOB_ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION_PCT		NUMBER(2,2)
MANAGER_ID		NUMBER(6)
DEPARTMENT_ID		NUMBER(4)

Departments

Name	Null?	Type

DEPARTMENT_ID	NOT NULL	NUMBER(4)
DEPARTMENT_NAME	NOT NULL	VARCHAR2(30)
MANAGER_ID		NUMBER(6)
LOCATION_ID		NUMBER(4)

- 1.The Hr department needs a report on job grades and salaries.To familiarize yourself with the SALGRADE table, first show the structure of the SALgrade table. Then create a query that displays the name,job,department name ,salary and grade for all employees.
- 2.Display the ENAME,SAL,MGR,DNAME of employees whose salary is more than 2000.Use the ON clause.
- 3.Display the EMPNO,ENAME,JOB,DEPTNO,DNAME,LOC of employees.Use left outer join.
- 4.Display the Ename, Dname of employees.Use right outer join.
- 5.Display the EMPNO,DNAME,LOC of employees.Use full outer join.

Set 14

Create table employees and department and solve the following queries

Name	Null?	Type
EMPLOYEE_ID	NOT NULL	NUMBER(6)
FIRST_NAME		VARCHAR2(20)
LAST_NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(25)
PHONE_NUMBER		VARCHAR2(20)
HIRE_DATE	NOT NULL	DATE
JOB_ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION_PCT		NUMBER(2,2)
MANAGER_ID		NUMBER(6)
DEPARTMENT_ID		NUMBER(4)

Name	Null?	Type
DEPARTMENT_ID	NOT NULL	NUMBER(4)
DEPARTMENT_NAME	NOT NULL	VARCHAR2(30)
MANAGER_ID		NUMBER(6)
LOCATION_ID		NUMBER(4)

1. Write a query to display the JOB, MGR, SAL, COMM, DNAME of employees whose JOB is SALESMAN.
2. The HR department needs a report of employees in Pune. Display the ENAME, job, DEPTNO and DNAME for all employees who work in Pune.
3. Display the Ename, Dname of all the employees, Also display those department name which do not have any employees working
4. Display the Ename, Dname of all the employees, Also display those department name which do not have any employees working
5. Display the EMPNO, ENAME, LOC of those employees who are working as CLERK

Set 15

Write a program that will prompt for a EMPNO, ENAME, SAL. If EMPNO does not exist in EMP table then INSERT the EMPNO, ENAME, SAL into the EMP table else UPDATE NAME, SAL for that EMPNO.

Set 16

Write a program that will prompt for number and display whether the number is ODD or EVEN number. Add user defined exception to check the number is negative.

Set 17

Write program that prompts for EMPNO and update his salary based on the below specification:
 If the employee belongs to DEPTNO 10 then update his salary by 10%.
 If the employee belongs to DEPTNO 20 then update his salary by 15%.
 If the employee belongs to other DEPTNO then update his salary with SALARY+COMM.

Set 18

Unnamed PL/SQL code block: Use of Control structure and Exception handling is mandatory. Write a PL/SQL block of code for the following requirements:-Consider table Stud(Roll, Att,Status) Write a PL/SQL block for following requirement and handle the exceptions. Roll no. of student will be entered by user. Attendance of roll no. entered by user will be checked in Stud table. If attendance is less than 75% then display the message "Term not granted" and set the status in stud table as "D". Otherwise display message "Term granted" and set the status in stud table as "ND"

Set 19

Unnamed PL/SQL code block: Use of Control structure and Exception handling is mandatory. Write a PL/SQL block of code for the following requirements:-
Schema:

1. Borrower(Rollin, Name, DateofIssue, NameofBook, Status)
2. Fine(Roll_no,Date,Amt)

Accept roll_no & name of book from user.

Check the number of days (from date of issue), if days are between 15 to 30 then fine amount will be Rs 5per day.

If no. of days>30, per day fine will be Rs 50 per day & for days less than 30, Rs. 5 per day.

After submitting the book, status will change from I to R.

If condition of fine is true, then details will be stored into fine table.

Set 20

Create following tables with primary and foreign key and solve the queries given below

student(S_ID,name,dept_name,tot_cred)
instructor(T_ID,name,dept_name,salary)
course(course_id,title,dept_name,credits)

- i. Find the average salary of instructor in those departments where the average salary is more than Rs. 42000/-.
- ii. Increase the salary of each instructor in the computer department by 10%.
- iii. Find the names of instructors whose names are neither 'Amol' nor 'Amit'.
- iv. Find the names of student which contains 'am' as its substring.
- v. Find the name of students from computer department that "DBMS" courses they take.

Set 21

Write a program that displays the Top N salaries using simple Explicit Cursors.

Set 22

PL/SQL Stored Procedure.

Write a Stored Procedure namely proc_Grade for the categorization of student. If marks scored by students in examination is <=1500 and marks>=990 then student will be placed in distinction category if marks scored are between 989 and 900 category is first class, if marks 899 and 825 category is Higher Second Class Write a PL/SQL block for using procedure created with above requirement. Stud_Marks(name, total_marks) Result(Roll,Name, Class).

Set 23

Create below tables, insert 5 records each and solve the following queries

Name	Null?	Type

EMPLOYEE_ID	NOT NULL	NUMBER(6)
FIRST_NAME		VARCHAR2(20)
LAST_NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(25)
PHONE_NUMBER		VARCHAR2(20)
HIRE_DATE	NOT NULL	DATE
JOB_ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION_PCT		NUMBER(2,2)
MANAGER_ID		NUMBER(6)
DEPARTMENT_ID		NUMBER(4)

SQL> desc departments;

Name	Null?	Type

DEPARTMENT_ID	NOT NULL	NUMBER(4)
DEPARTMENT_NAME	NOT NULL	VARCHAR2(30)
MANAGER_ID		NUMBER(6)
LOCATION_ID		NUMBER(4)

1. List employees whose salary is greater than the salary of ALLEN.
2. Find employees whose job is the same as that of MARTIN.
3. Display employees who are not working in the department of KING.
4. Find employees who earn more than the salary of BLAKE.
5. Display employees hired after JONES.
6. Find employees whose department number is the same as CLARK.

Set 24

Create below tables employee and department, insert 5 records each and solve the following queries

Name	Null?	Type

EMPLOYEE_ID	NOT NULL	NUMBER(6)
FIRST_NAME		VARCHAR2(20)
LAST_NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(25)
PHONE_NUMBER		VARCHAR2(20)
HIRE_DATE	NOT NULL	DATE
JOB_ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION_PCT		NUMBER(2,2)
MANAGER_ID		NUMBER(6)
DEPARTMENT_ID		NUMBER(4)

Name	Null?	Type

DEPARTMENT_ID	NOT NULL	NUMBER(4)
DEPARTMENT_NAME	NOT NULL	VARCHAR2(30)
MANAGER_ID		NUMBER(6)
LOCATION_ID		NUMBER(4)

1. Find employees whose salary is greater than ANY employee in department 30.
2. Display employees whose salary is greater than ALL employees in department 20.
3. List employees working in departments where the job is MANAGER.
4. Find employees whose salary is greater than the average salary of their own department.
5. Display employees who earn the highest salary in their department.
6. Find employees who have the same job as at least one employee in department 20.

Set 25

Write a database trigger on Library table. The System should keep track of the records that are being updated or deleted. The old value of updated or deleted records should be added in Library_Audit table. Frame the problem statement for writing Database Triggers of all types, in-line with above statement. The problem statement should clearly state the requirements.

Set 26

Collection "orderinfo" which contains the documents given as below

```
{
  cust_id:123
  cust_name:"abc",
  status:"A",
  price:250
}
```

- 1) find the average price for each customers having status 'A'
- 2) Display the status of the customers whose amount/price lie between 100 and 1000
- 3) Display the customers information without "_id" .
- 4) create a simple index on orderinfo collection and fire the queries.

Set 27

Collection "movies" which contains the documents given as below(Perform on Mongo Terminal)

```
{
  name: "Movie1",
  type: "action",
  budget: 1000000
  producer: {
    name: "producer1",
    address: "PUNE"
  }
}
```

- i. Find the name of the movie having budget greater than 1,00,000.
- ii. Find the name of producer who lives in Pune
- iii. Update the type of movie "action" to "horror"
- iv. Find all the documents produced by name "producer1" with their address

Set 28

Consider following structure for MongoDB collection and write a query for following requirements in MongoDB

Teachers (Tname,dno,Experience,Salary,Date_of_Joining)

Department (Dno,Dname)

Students(Sname,Roll_No,Class)

1. Write a query to create above collection insert some sample documents.
2. Find the information about all teachers of Dno=2 and having salary greater than or equal to 10,000/-
3. Find the student information having Roll_no=2 or Sname='xyz'
4. Update student name whose Roll_No=5
5. Delete all student whose Class is 'FE'
6. Apply index on Students Collection

Set 29

Perform aggregation and Indexing using MongoDB on below database

1. Create a database department
2. Create a collection as teacher with fields as name , department ,experience and salary
3. Display the department wise average salary.
4. Display the no. Of employees working in each department.
5. Display the department wise minimum salary.
6. Apply index and drop index

Set 30

1. Create database product_detail
2. Show Database
3. Create collection product
4. Show collection
5. Insert below documents

```
{ "item": "pen", "price": 10, "quantity": 2 } { "item": "book", "price": 50, "quantity": 1 } { "item": "pen", "price": 10, "quantity": 5 } { "item": "pencil", "price": 5, "quantity": 10 }
```
6. Retrieve documents from collection.
7. Create index on item in ascending order
8. Show index
9. Create index on price in ascending and quantity as descending
10. Drop index on price

Set 31

Collection "orderinfo" which contains the documents given as below

```
{
  cust_id:123
  cust_name:"abc",
  status:"A",
  price:250
}
```

- i. Display the name of the customer having the price between 250 and 450
- ii. Increment the price by 10 for cust_id: 123 and decrement the price by 5 for cust_id: 124
- iii. Remove any one of the field from the orderinfo collection.
- iv. Find the name of the customer whose status is either A or price is 250 or both.

Set 32

Emp(emp_id,ename, street, city)

works(emp_id,c_id,ename, cname, sal)

Company(c_id,cname, city)

Manager(mgr_id, mgrname)

- i. Modify the database so that a particular company (eg. ABC) now in Pune
- ii. Give all managers of Mbank a 10% raise. If salary is >20,000, give only 3% raise.
- iii. Find out the names of all the employees who works in 'Bosch' company in city Pune
- iv. Delete all records in the works table for employees of a particular company (Eg, SBC Company) whose salary>50,000.

Set 33

1. Create database Employee.
- 2.Create collection emp1 using createCollection method.
- 3.Insert 4 to 5 documents in emp1 collection.(eno,ename,address,sal)
- 4.display all documents.
- 5.Display all employess having salary greater than 5000
- 6.Display all employess having salary less than 15000
- 7.Display all employess having salary greater than 10000 and less than 20000.
- 8.Update sal of all employee with 10%
- 9.Delete employee having salary less than 5000.
- 10.Rename collection emp1 with employee1.
- 11.Display employee whose name start with n.
- 12.sort name in ascending order.
- 13.Create a new database with name Employee1.
- 14.Drop employee1 database.
- 15.Create collection emp1 using insert method.
- 16.Drop collection emp1.

Set 34

Employee(employee_name,street,city)
Works(employee_name,company_name,salary)
Company(company_name,city)
Manages(employee_name,manager_name)

1. Find the names of employees who work for First Bank Corporation.
2. Find the names and cities of residence of all employees who work for First Bank Corporation
3. Find the names, street addresses, and cities of residence of all employees who work for First Bank Corporation and earn more than \$10000.
4. Find all employees in the database who live in the same cities as the companies for which they work.
5. Find all employees in the database who live in the same cities and on the same streets as do their manager.
6. Find all employees in the database who do not work for First Bank Corporation
7. Give all employees of First Bank Corporation a 10% raise
8. Delete all tuples in the "Works" relation for employees of "Small Bank Corporation".

Set 35

Employee(employee_name,street,city)
Works(employee_name,company_name,salary)
Company(company_name,city)
Manages(employee_name,manager_name)

- 1) Find all employees in the database who earn more than each employee of Small Bank Corporation
- 2) Assume that the company may be located in several cities. Find all companies located in every city in which Small Bank Corporation is located.
- 3) Find all employees who earn more than the average salary of all employees of their companies.
- 4) Find the company that has the most employees.
- 5) Find the company that has the smallest payroll.
- 6) Find those companies whose employees earn a higher salary, on average, than the average salary at First Bank Corporation.
- 7) Modify the database so that "Jones" now lives in Newtown.